

MD KAYSER AHMED HRIDOY

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with a research background in biomedical image analysis and deep learning, with contributions in retinal and multimodal medical imaging. My work focuses on developing segmentation and classification pipelines for retinal and MRI data under real-world clinical constraints, integrating CNN and graph-based approaches. I am interested in clinically deployable AI systems, particularly in medical image segmentation and graph-based learning, with an emphasis on robustness, reproducibility, and handling imperfect medical data.

EDUCATION

University of Antwerp

Master of Science – Electronics & ICT Engineering Technology

Antwerp, Belgium

Expected: September 2026

- Incoming graduate student; admission secured for the academic year 2026–2027.

National Institute of Technology Silchar (NIT Silchar)

Bachelor of Technology – Electronics & Communication Engineering

Assam, India

2021 – 2025

- CGPA: 6.74/10.00 (First Class) | 3.38/4.00 (WES iGPA) | 200 Credits (Indian Credit System)
- **Undergraduate Thesis:** Detection of Cardiovascular Diseases through Retinal Fundus Images using Deep Learning
 - * Group Leader – responsible for research design, experimental implementation, and coordination
 - * Awarded full marks (100%) for thesis evaluation
- **Relevant Coursework:** Programming, Data Structures & Algorithms, Signals & Systems, Digital Signal Processing, Machine Learning, Deep Learning & Applications, Mathematics, Statistics & Random Process, Digital Electronics, Wireless Sensor Networks

RESEARCH EXPERIENCE

Research Intern – Biomedical Imaging Laboratory

Department of Electronics & Communication Engineering

NIT Silchar, India

July 2025 – December 2025

- Designed a deep learning pipeline for diabetic retinopathy analysis from retinal fundus images, focusing on microaneurysm-level feature representation and classification.
- Conducted systematic experiments on CNN-based models, evaluating performance across architectures and assessing robustness under heterogeneous imaging conditions.
- Utilized multi-source retinal datasets and contributed to experimental validation and manuscript preparation for IEEE and Elsevier-indexed publications.

Research Intern – SERB SSR Project (Government of India)

Science and Engineering Research Board, Department of ECE

NIT Silchar, India

May 2025 – June 2025

- Designed deep learning pipelines for MRI-based Alzheimer's disease risk prediction, including ResAttU-Net segmentation and CNN-based classification using structural and functional MRI data.
- Performed multimodal MRI preprocessing, grey matter extraction, data integration, and performance evaluation.
- Co-authored a Q1 journal manuscript submitted to *Biomedical Signal Processing and Control* (Elsevier), under supervision of Dr. Tripti Goel.

PUBLICATIONS

Published / Accepted

- [1] S. R. Barman, **Md Kayser Ahmed Hridoy**, R. Murugan, "RGAT: Retinal Graph Attention Networks with Super-pixel Based Feature Hierarchies for CVD Associated Retinal Phenotype Classification," *Pattern Recognition Letters*, **Q1 Elsevier**. DOI: 10.1016/j.patrec.2026.04.027
- [2] **Md Kayser Ahmed Hridoy**, S. R. Barman, R. Murugan, "Classification of Cardiovascular Diseases through Retinal Fundus Images Using Deep Learning," *Proc. 14th IEEE Int. Conf. Communication Systems and Network Technologies (CSNT)*, VIT Bhopal, India, Mar. 2025. *Scopus Indexed*. DOI: 10.1109/CSNT64827.2025.10969053

- [3] D. Deb, **Md Kayser Ahmed Hridoy**, S. R. Barman, R. Murugan, “Robust Microaneurysm Detection in Retinal Fundus Images Using an Optimized ResNet50-UNet Model,” *Proc. 2nd Int. Conf. Signal Processing and Computer Vision (SIPCOV)*, NIT Silchar, India, Aug. 2025. DOI: 10.1007/978-3-032-15809-3_16
- [4] A. K. Singh, M. R. I. Midul, M. S. Mia, P. K. Gupta, **Md Kayser Ahmed Hridoy**, C. H. Rahaman, R. H. Laskar, “Edge-Guided U-Net with Frequency Domain Refinement for Efficient Skin Lesion Segmentation,” in *Proceedings of the IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, Nov. 2025. doi: 10.1109/BECITHCON69222.2025.11504184.

Under Review

- [5] **Md Kayser Ahmed Hridoy**, S. R. Barman, R. Murugan, “Integrating Multi-Feature Retinal Maps from Fundus Images for CVD-Associated Retinal Phenotypes Classification Using Deep Learning,” *Image and Vision Computing*, **Q1 Elsevier**. [Under Review]
- [6] D. Deb, **Md Kayser Ahmed Hridoy**, S. R. Barman, R. Murugan, S. A. Ananna, “MAGIC-DR: Microaneurysm Graph-Based Identification and Classification for Diabetic Retinopathy,” *IEEE Multimedia*, **Q2**. [Under Review]
- [7] S. Barman, **Md Kayser Ahmed Hridoy**, S. Goswami, T. Goel, “ResAttUNet and Multi-Stage Classifier Network for Gray Matter Segmentation, Quantification, and Alzheimer’s Disease Prediction from MRI Images,” *Biomedical Signal Processing and Control*, **Q1 Elsevier**. [Under Review]
- [8] D. Deb, R. Murugan, **Md Kayser Ahmed Hridoy**, S. R. Barman, “FD-GNet: A Feature Driven Graph Neural Network for Early Classification of Retinal Abnormalities,” *IEEE Access*, **Q1**. [Under Review]

PROJECTS

TransAttUnet++: Attention-Guided U-Net with Transformer for Image Segmentation

| *PyTorch, Python*

Nov 2024 – Jan 2025

- Designed a hybrid U-Net + Transformer self-attention segmentation model with frequency-domain attention for medical image analysis, integrated multi-scale skip connections and residual pathways.
- Achieved 88.14% Dice coefficient and 95.32% accuracy, outperforming standard U-Net baselines.
- Implemented a full PyTorch training pipeline with combined loss functions, mixed-precision training, learning-rate scheduling, and model checkpointing. [GitHub](#)

A DenseNet Framework for Lung Cancer Classification from OCT Images

| *TensorFlow, Keras*

Feb 2024 – Apr 2024

- Developed a DenseNet169-based multiclass classification model for lung OCT images (benign, malignant, normal) with robust data augmentation and standard performance metrics.
- Designed an interactive Tkinter GUI for real-time inference and visualization, enabling deployment for non-technical end users.

ACADEMIC SERVICE

Peer Reviewer – *Biomedical Signal Processing and Control*, Elsevier (Q1 Journal)

2025–2026

HONOURS & AWARDS

Best Project (Thesis) Award – Department of ECE, NIT Silchar

May 2025

- Awarded for thesis titled “Detection of Cardiovascular Diseases through Fundus Images Using Deep Learning.” Recognized for methodological rigor, originality, and experimental validation; resulted in three publications. Ranked **1st in AI Track** and 4th overall among 65 competing groups.

ICCR Scholarship (Government of India) – Indian Council for Cultural Relations

2021–2025

- Awarded through highly competitive merit-based selection to pursue a Bachelor’s degree in Engineering in India, covering tuition fees and living expenses for the full duration of study.

PROFESSIONAL MEMBERSHIPS

Student Member, Institute of Electrical and Electronics Engineers (IEEE), USA

Membership ID: 102046485

LANGUAGE PROFICIENCY

Bengali: Native

English: Professional Proficiency – IELTS Academic Overall Band 6.5

Hindi: Conversational

TECHNICAL SKILLS

Programming Languages: Python, C

Deep Learning Frameworks: PyTorch, TensorFlow, Keras, TorchVision

ML / Computer Vision Libraries: PyTorch Geometric (PyG), Scikit-learn, OpenCV

Graph Learning: PyTorch Geometric, NetworkX

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Development Platforms: Jupyter Notebook, Google Colab, VS Code, Overleaf

Version Control & Collaboration: GitHub, Google Workspace, Microsoft Office Suite, Canva Pro

Soft Skills: Analytical Thinking, Research Orientation, Problem Solving, Teamwork, Technical Communication, Time Management

CERTIFICATIONS

A Complete Guide to the Key Components of a Research Article – Wiley Researcher Academy

Structured training on research article structure, scientific writing, peer review, and publication ethics.

Mastering Data Structures & Algorithms using C and C++ – Udemy

Fundamental data structures and algorithms including arrays, linked lists, stacks, queues, trees, graphs, and algorithmic problem-solving.

EXTRACURRICULAR ACTIVITIES

Convener – Prothom Alo Bondhushova, NIT Silchar Subsection

2024–2025

Led and coordinated student-led cultural and social initiatives; oversaw event planning, team coordination, and community engagement.

Volunteer – Technoosis Annual Technical Festival, NIT Silchar

2023–2024

Organized “School Genius,” an outreach event engaging 200+ school students in STEM activities.

Member – Machine Learning Club, NIT Silchar

2022–2025

Participated in technical discussions and learning activities related to machine learning and artificial intelligence.

REFERENCES

Dr. R. Murugan

Assistant Professor

Dept. of Electronics & Communication Engineering

National Institute of Technology Silchar

Assam – 788010, India

murugan.rmn@ece.nits.ac.in

Phone: +91 96000 81925

(Thesis & Research Supervisor)

Dr. Tripti Goel

Assistant Professor, Grade-I

Dept. of Electronics & Communication Engineering

National Institute of Technology Silchar

Assam – 788010, India

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Phone: +91 95413 45413

(SERB-SSR Research Supervisor)